QFT

The Quantum Fourier Transform

A step-by-step chapter: from roots of unity to Shor's algorithm

What you will be able to do after this chapter

- compute sums, products and phases of roots of unity quickly and without a calculator;
- manipulate qubit states, measurements, gates and tensor products with confidence;
- write down the QFT of any small state, both from the definition and from the product formula;
- build, count and invert the QFT circuit, and trace it gate by gate;
- analyse periodic states, quantum phase estimation and Shor's algorithm, including their failure modes;
- reason about the approximate and semiclassical QFT and the algorithms that use the QFT;
- solve every type of question in the accompanying 80-question QFT assessment.

Companion material in the same repository: `presentations/QFT_Lecture.pdf` (slides), `QFT.ipynb` (CUDA-Q notebook) and `assessments/` (question paper and answer book).

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QFT — The Quantum Fourier Transform

How to use this chapter

The chapter is built as a staircase. Sections 1–4 are the *mathematics* you need (complex numbers, binary fractions, matrices, the discrete Fourier transform). Sections 5–6 are the *quantum mechanics* (states, measurement, gates). Sections 7–11 are the *QFT itself*. Sections 12–14 are the *advanced* material: phase estimation, Shor's algorithm, approximate QFT and applications. Section 15 collects problem-solving strategies and Section 16 has practice problems with answers.

Each section has: the idea in words, the precise statement, a derivation, **worked examples** solved step by step, **common traps** (the wrong answers examiners like to offer) and a short checklist. Do the worked examples with a pencil before reading the solution.

Definition: Conventions used everywhere in this chapter

- n qubits, $N = 2^n$, $\omega = \omega_N = e^{2\pi i/N}$.
- $|x\rangle = |x_1 x_2 \dots x_n\rangle$ with $x = x_1 2^{n-1} + x_2 2^{n-2} + \cdots + x_n 2^0$: **qubit 1 is the most significant bit**.
- $\mathrm{QFT}\,|x\rangle = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega^{xy} |y\rangle$ and $\mathrm{QFT}^{-1}\,|y\rangle = \frac{1}{\sqrt{N}} \sum_x \omega^{-xy} |x\rangle$.
- $R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$; so $R_1 = Z$, $R_2 = S$, $R_3 = T$.
- Two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; log means $\log_2$.

1 Complex numbers and roots of unity

1.1 Complex numbers as arrows

A complex number $z = a + ib$ is an arrow in the plane. In polar form $z = re^{i\theta}$ with $r = |z|$ and angle θ. Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ gives the three facts we use all the time:

$$|e^{i\theta}| = 1, \qquad e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}, \qquad \overline{e^{i\theta}} = e^{-i\theta} = \frac{1}{e^{i\theta}}.$$

So *multiplying by* $e^{i\theta}$ *rotates by* θ and never changes the length. Also $e^{2\pi ik} = 1$ for every integer k: a whole number of turns does nothing. This single fact will explain most of the QFT.

Useful values: $e^{i\pi/2} = i$, $e^{i\pi} = -1$, $e^{3i\pi/2} = -i$, $e^{i\pi/4} = \frac{1+i}{\sqrt{2}}$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$.

Worked example 1.1: Reducing a large phase

Find $e^{2\pi i \cdot 29/8}$.

Step 1. Split off whole turns: $\frac{29}{8} = 3 + \frac{5}{8}$.

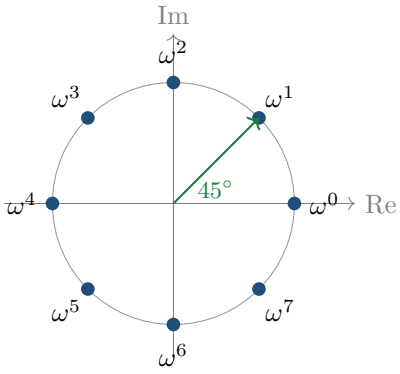
Step 2. $e^{2\pi i \cdot 3} = 1$, so $e^{2\pi i \cdot 29/8} = e^{2\pi i \cdot 5/8} = e^{i\,5\pi/4} = -\frac{1+i}{\sqrt{2}}$.

Rule: only the *fractional part* of the number of turns matters.

1.2 The N-th roots of unity

Definition: Roots of unity

The solutions of $z^N = 1$ are $1, \omega, \omega^2, \ldots, \omega^{N-1}$ with $\omega = e^{2\pi i/N}$. They are the vertices of a regular N-gon on the unit circle. Exponents may be reduced modulo N: $\omega^k = \omega^{k \bmod N}$, and $\omega^{-k} = \overline{\omega^k} = \omega^{N-k}$.



The eight 8th roots of unity, $\omega = e^{2\pi i/8} = e^{i\pi/4}$. Note $\omega^2 = i$, $\omega^4 = -1$, $\omega^6 = -i$.

1.3 Geometric series — the most important formula

For any $z \neq 1$,

$$\sum_{k=0}^{M-1} z^k = 1 + z + \cdots + z^{M-1} = \frac{1 - z^M}{1 - z}.$$

Proof: call the sum S; then $zS = z + \cdots + z^M$, so $S - zS = 1 - z^M$.

Key result: Orthogonality of roots of unity

$$\sum_{j=0}^{N-1} \omega^{jk} = \begin{cases} N & \text{if } N \mid k, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. If $N \mid k$ every term is 1. Otherwise $z = \omega^k \neq 1$ and $z^N = e^{2\pi i k} = 1$, so the geometric series gives $\frac{1-1}{1-z} = 0$.

Geometrically: the arrows ω^{jk} are spread symmetrically round the circle and cancel. This is *destructive interference*; when they all point the same way ($N \mid k$) we get *constructive interference*.

Common trap

"The numbers ω^{jk}, $j = 0, \dots, N-1$, are always N *distinct* vertices." False when $g = \gcd(k, N) > 1$: for $N = 8$, $k = 2$ you get $1, i, -1, -i$ twice. The sum is still 0 (a square traversed twice), but the "distinct vertices" explanation is wrong. Assertion–Reason questions love this distinction.

Worked example 1.2: Counting pairs with a non-zero sum

How many pairs (x, y) with $x, y \in \{0, \dots, 7\}$ make $\sum_{z=0}^{7} e^{2\pi i(x+y)z/8} \neq 0$?

Step 1. By orthogonality the sum is non-zero exactly when $8 \mid (x + y)$.

Step 2. $x + y \in \{0, \dots, 14\}$, so we need $x + y = 0$ or $x + y = 8$.

Step 3. $x + y = 0$: only $(0, 0)$. $x + y = 8$: $x = 1, \dots, 7$ (7 pairs). Total 8.

Shortcut: for each x there is exactly one $y \equiv -x \pmod 8$, so the answer is always N.

1.4 Weighted sums: differentiate the geometric series

Differentiating $\sum_{k=0}^{M-1} z^k = \frac{1-z^M}{1-z}$ and multiplying by z gives

$$\sum_{k=0}^{M-1} k z^k = \frac{z(1 - Mz^{M-1} + (M-1)z^M)}{(1-z)^2}.$$

Worked example 1.3: $\sum_k k\,\omega^k$ for $N = 4$

Let $\omega = i$ ($N = 4$). Directly: $0 + i + 2(-1) + 3(-i) = -2 - 2i$. Formula with $z = i$, $M = 4$, $z^4 = 1$, $z^3 = -i$: $\frac{i(1-4(-i)+3)}{(1-i)^2} = \frac{i(4+4i)}{-2i} = -2 - 2i$. ✓ In general, for $z = \omega_N$: $\sum_{k=0}^{N-1} k\,\omega^k = \frac{\omega(1-N\omega^{-1}+N-1)}{(1-\omega)^2} = \frac{N(\omega-1)}{(1-\omega)^2} = \frac{N}{\omega-1}$. (Check: $\frac{4}{i-1} = \frac{4(-i-1)}{2} = -2 - 2i$ ✓.)

1.5 Products of roots of unity

Since the roots of $z^N - 1$ are exactly $\omega^0, \dots, \omega^{N-1}$:

$$z^N - 1 = \prod_{k=0}^{N-1} (z - \omega^k) \quad \Longrightarrow \quad 1 + z + \cdots + z^{N-1} = \prod_{k=1}^{N-1} (z - \omega^k).$$

Putting $z = 1$: $\prod_{k=1}^{N-1}(1-\omega^k) = N$, hence $\prod_{k=1}^{N-1} |1-\omega^k| = N$. *Example:* $N = 4$: $|1-i|\,|1+1|\,|1+i| = \sqrt{2} \cdot 2 \cdot \sqrt{2} = 4$. ✓

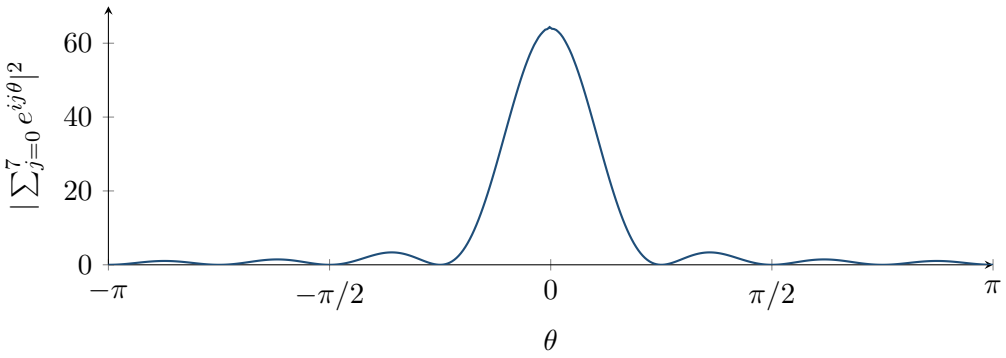
1.6 Interference patterns: $|\sum e^{ij\theta}|^2$

Key result: The Fourier peak shape

$$\left| \sum_{j=0}^{K-1} e^{ij\theta} \right| = \left| \frac{\sin(K\theta/2)}{\sin(\theta/2)} \right|, \qquad \left| \sum_{j=0}^{K-1} e^{ij\theta} \right|^2 = \frac{\sin^2(K\theta/2)}{\sin^2(\theta/2)}.$$

Maximum K^2 at $\theta = 0$ (all arrows aligned); zeros at $\theta = \frac{2\pi m}{K}$, $m = 1, \dots, K-1$.

Derivation: $\sum_j e^{ij\theta} = \frac{1-e^{iK\theta}}{1-e^{i\theta}}$ and $1 - e^{i\phi} = e^{i\phi/2}(e^{-i\phi/2} - e^{i\phi/2}) = -2i\, e^{i\phi/2} \sin(\phi/2)$, so the ratio has modulus $|\sin(K\theta/2)/\sin(\theta/2)|$.



$K = 8$: a tall central peak of height 64, first zeros at $\pm 2\pi/8 = \pm\pi/4$, small side lobes. Every QFT output peak has this shape.

Worked example 1.4: Maximum and first zero

For $f(\theta) = |1 + e^{i\theta} + e^{2i\theta}|^2$: $K = 3$, maximum 9, first positive zero at $\theta = 2\pi/3$ (three arrows at 120° cancel). Check: $1 + e^{2\pi i/3} + e^{4\pi i/3} = 0$. ✓

1.7 Gauss sums (traces of Fourier matrices)

$\sum_{x=0}^{N-1} \omega^{x^2}$ is a *Gauss sum*. For small N just list the squares mod N. *Example* $N = 8$*:* $x^2 \bmod 8 = 0, 1, 4, 1, 0, 1, 4, 1$, so the sum is $2 \cdot 1 + 4\omega + 2\omega^4 = 2 + 4\omega - 2 = 4\omega$. *($\omega^4 = -1$.)*

Checklist 1. Reduce exponents mod N; use the geometric series; for sums of roots of unity first ask "does N divide the exponent?"; remember |phase| = 1.

2 Binary numbers and binary fractions

2.1 Integers in binary

$x = x_1 x_2 \ldots x_n$ (qubit 1 first) means $x = \sum_{j=1}^{n} x_j 2^{n-j}$. Example: $1101_2 = 8 + 4 + 0 + 1 = 13$.

Definition: Binary fraction

$0.x_l x_{l+1} \ldots x_m = \frac{x_l}{2} + \frac{x_{l+1}}{4} + \cdots + \frac{x_m}{2^{m-l+1}}$. Examples: $0.1_2 = \frac{1}{2}$, $0.01_2 = \frac{1}{4}$, $0.11_2 = \frac{3}{4}$, $0.101_2 = \frac{5}{8}$, $0.011_2 = \frac{3}{8}$.

2.2 Shifting the binary point

Dividing by 2^l moves the binary point l places to the left; multiplying moves it right:

$$\frac{x}{2^l} = (\text{integer}) + 0.x_{n-l+1} \ldots x_n, \qquad 2^k \cdot 0.\varphi_1\varphi_2\varphi_3 \cdots = \varphi_1 \ldots \varphi_k.\varphi_{k+1}\varphi_{k+2} \ldots$$

Key result: Only the last bits survive in a phase

$$e^{2\pi i\, x/2^l} = e^{2\pi i\, 0.x_{n-l+1} \ldots x_n}, \qquad e^{2\pi i\, 2^k \varphi} = e^{2\pi i\, 0.\varphi_{k+1}\varphi_{k+2} \ldots}$$

because the integer part is a whole number of turns.

Worked example 2.5: A phase from a 6-bit number

$x = 27 = 011011_2$. Find $e^{2\pi i x/2^3}$.

Step 1. $27/8 = 011.011_2 = 3 + 0.011_2 = 3 + \frac{3}{8}$.

Step 2. Drop the 3 turns: $e^{2\pi i \cdot 3/8} = e^{3\pi i/4}$.

Only the last three bits 011 mattered.

Worked example 2.6: The bit carried by U^{2^k}

$\varphi = 0.\varphi_1\varphi_2\varphi_3$. Then $e^{2\pi i\, 2\varphi} = e^{2\pi i\, 0.\varphi_2\varphi_3}$ and $e^{2\pi i\, 4\varphi} = e^{2\pi i\, 0.\varphi_3} = (-1)^{\varphi_3}$. So in phase estimation the qubit controlling U^4 carries only the last bit. (Recall $e^{2\pi i \cdot \frac{1}{2}} = -1$.)

2.3 Fractions that never terminate

$\frac{p}{q}$ (in lowest terms) has a terminating binary expansion iff q is a power of 2. So $\frac{1}{3} = 0.010101\ldots_2$ and $\frac{1}{5} = 0.00110011\ldots_2$ never terminate: *no finite register can represent them exactly.* Long division in base 2 for $\frac{1}{3}$: $1 \cdot 2 = 2 < 3 \to 0$; $2 \cdot 2 = 4 \geq 3 \to 1$, remainder 1; repeat $\Rightarrow 0.\overline{01}$.

3 Linear algebra for quantum computing

3.1 Vectors, inner products, unitary matrices

$\langle u|v\rangle = \sum_k \overline{u_k} v_k$ (conjugate the *first* vector). $\|v\|^2 = \langle v|v\rangle$. A matrix U is **unitary** if $U^\dagger U = I$, where $U^\dagger = \overline{U}^T$. Equivalent tests:

- the columns are orthonormal (each has length 1, different columns have inner product 0);
- equivalently the rows are orthonormal;
- U preserves lengths: $\|Uv\| = \|v\|$ for all v.

Consequences: $U^{-1} = U^\dagger$; $|\det U| = 1$; every eigenvalue has $|\lambda| = 1$. *Proof of* $|\det U| = 1$*:* $|\det U|^2 = \overline{\det U} \det U = \det(U^\dagger U) = \det I = 1$.

Worked example 3.7: Completing a unitary matrix

Find a, b such that $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & a \\ i & b \end{pmatrix}$ is unitary with $a = 1$.

Step 1. Columns $c_1 = \frac{1}{\sqrt{2}}(1, i)$, $c_2 = \frac{1}{\sqrt{2}}(1, b)$.

Step 2. $\langle c_1|c_2\rangle = \frac{1}{2}(\bar{1} \cdot 1 + \bar{i}\, b) = \frac{1}{2}(1 - ib) = 0 \Rightarrow b = \frac{1}{i} = -i$.

Step 3. Check $\|c_2\|^2 = \frac{1}{2}(1 + |-i|^2) = 1$. ✓ So $V = \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ i & -i \end{smallmatrix}\right)$.

Do not forget the complex conjugate in the inner product — it is the most common error.

Common trap

A matrix with two *equal* (or proportional) columns, such as $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ i & i\end{smallmatrix}\right)$, has determinant 0 and can never be a gate: it would send two different inputs to the same output and destroy information.

3.2 Eigenvalues from a polynomial identity

If $U^m = I$ and $Uv = \lambda v$ then $v = U^m v = \lambda^m v$, so $\lambda^m = 1$. Example: $Z^2 = I \Rightarrow \lambda = \pm 1$. The trace is the sum and the determinant the product of the eigenvalues — use these to identify which roots occur.

3.3 Tensor products

For $a = (a_0, a_1)$, $b = (b_0, b_1)$: $a \otimes b = (a_0b_0, a_0b_1, a_1b_0, a_1b_1)$ in the order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. For matrices, $(A \otimes B)(a \otimes b) = Aa \otimes Bb$ and $A \otimes B = \left(\begin{smallmatrix}a_{11}B & a_{12}B\\ a_{21}B & a_{22}B\end{smallmatrix}\right)$. *Example:* $I \otimes Z = \mathrm{diag}(1, -1, 1, -1)$ and $Z \otimes I = \mathrm{diag}(1, 1, -1, -1)$ — the position matters.

3.4 Commuting matrices

Diagonal matrices always commute with each other. A permutation P commutes with a diagonal D iff D has equal entries on the positions that P swaps. A gate that mixes two basis states (like H) commutes with a diagonal D only if D has equal entries on the states it mixes.

Checklist 3. Conjugate in inner products; test unitarity by columns; use $U^m = I$ to get eigenvalues; respect the order in tensor products.

4 The discrete Fourier transform

4.1 Definition and matrix

Definition: Unitary DFT

For a vector $\alpha = (\alpha_0, \ldots, \alpha_{N-1})$: $\beta_y = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \alpha_x \, \omega^{xy}$, i.e. $\beta = F\alpha$ with $F_{yx} = \frac{\omega^{xy}}{\sqrt{N}}$.

For $N = 2$: $F = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right) = H$. For $N = 4$ ($\omega = i$):

$$F_4 = \frac{1}{2}\begin{pmatrix}1 & 1 & 1 & 1\\ 1 & i & -1 & -i\\ 1 & -1 & 1 & -1\\ 1 & -i & -1 & i\end{pmatrix}.$$

Memory aid: row y, column x holds i^{xy}; row 3 is $i^0, i^3, i^6, i^9 = 1, -i, -1, i$.

F is unitary: $(F^\dagger F)_{x'x} = \frac{1}{N}\sum_y \omega^{y(x-x')} = \delta_{xx'}$ by orthogonality.

4.2 Parseval and special inputs

Key result: Parseval

$\sum_y |\beta_y|^2 = \sum_x |\alpha_x|^2$ (unitarity). For the *un-normalised* transform $A_y = \sum_x a_x \omega^{xy} = \sqrt{N}\,\beta_y$: $\sum_y |A_y|^2 = N \sum_x |a_x|^2$.

- $\beta_0 = \frac{1}{\sqrt{N}} \sum_x \alpha_x$ (the "average" or DC component);
- a delta $\alpha = e_x$ gives $|\beta_y| = \frac{1}{\sqrt{N}}$ for every y (flat spectrum);
- a constant vector gives a delta at $y = 0$.

Worked example 4.8: All four coefficients and a sanity check

$\alpha = \frac{1}{\sqrt{10}}(1, 0, 3, 0)$, $N = 4$, so $\beta_y = \frac{1}{2} \sum_x \alpha_x i^{xy}$ and only $x = 0, 2$ contribute: $\beta_y = \frac{1+3\,i^{2y}}{2\sqrt{10}} = \frac{1+3(-1)^y}{2\sqrt{10}}$.

Step 1. $y = 0, 2$: $\beta_y = \frac{4}{2\sqrt{10}} = \frac{2}{\sqrt{10}}$, probability $\frac{4}{10}$ each.

Step 2. $y = 1, 3$: $\beta_y = \frac{-2}{2\sqrt{10}} = -\frac{1}{\sqrt{10}}$, probability $\frac{1}{10}$ each.

Step 3. Parseval: $\frac{4+4+1+1}{10} = 1$. ✓

4.3 Shift theorem

Key result: Shift ↔ phase ramp

If $\alpha'_x = \alpha_{x-s}$ (indices mod N), then $\beta'_y = \omega^{sy}\beta_y$. Magnitudes $|\beta_y|$ are unchanged.

Proof: $\beta'_y = \frac{1}{\sqrt{N}} \sum_x \alpha_{x-s}\omega^{xy} = \frac{1}{\sqrt{N}} \sum_u \alpha_u \omega^{(u+s)y} = \omega^{sy}\beta_y$ $(u = x - s)$.

4.4 Periodic sequences: peaks

Key result: Period r dividing N

If α is $\frac{1}{\sqrt{N/r}}$ on $x_0, x_0 + r, x_0 + 2r, \ldots$ and 0 elsewhere, and $r \mid N$, then $\beta_y = \frac{1}{\sqrt{r}}\,\omega^{x_0 y}$ for $y \in \{0, \frac{N}{r}, \frac{2N}{r}, \ldots\}$ and 0 otherwise: exactly r peaks, each with probability $\frac{1}{r}$.

Derivation: with $K = N/r$ terms, $\beta_y = \frac{\omega^{x_0 y}}{\sqrt{NK}} \sum_{j=0}^{K-1} \omega^{rjy}$ and $\omega^{ry} = e^{2\pi i\, y/K}$; the sum is K if $K \mid y$ and 0 otherwise. Then $|\beta_y| = \frac{K}{\sqrt{NK}} = \sqrt{K/N} = \frac{1}{\sqrt{r}}$.

Worked example 4.9: Period 4 in 16

$\alpha = \frac{1}{2}(|1\rangle + |5\rangle + |9\rangle + |13\rangle)$ $(N = 16, r = 4, x_0 = 1)$. Peaks at $y = 0, 4, 8, 12$ with probability $\frac{1}{4}$ each; the phase of β_4 is $\omega^{1\cdot 4} = e^{2\pi i\, 4/16} = i$, i.e. 90°. Halving the period doubles the spacing: period r in x ⇔ spacing N/r in y.

4.5 When the period does not divide N: leakage

If $r \nmid N$ the number of terms K is $\lfloor N/r \rfloor$ or $\lceil N/r \rceil$ and the peaks sit near (not at) the non-integer points kN/r. Using the peak-shape formula with $\theta = 2\pi ry/N$:

$$|A_y| = \left| \frac{\sin(\pi K r y/N)}{\sin(\pi r y/N)} \right|.$$

The peak is **highest when ry/N is closest to an integer**.

Worked example 4.10: Which y gives the largest peak?

$N = 16$, $a_x = 1$ for $x \equiv 0 \pmod 3$ ($x = 0, 3, \ldots, 15$: $K = 6$). Near $y \approx 16k/3 = 5.33, 10.67$: $3 \cdot 5/16 = 0.9375$ (distance 0.0625 from 1) and $3 \cdot 11/16 = 2.0625$ (distance 0.0625), $3 \cdot 6/16 = 1.125$ (distance 0.125). So $y = 5$ and $y = 11$ tie for the largest peak: $|A_5| = |\sin(6 \cdot 0.9375\pi)/\sin(0.9375\pi)| = 0.9239/0.1951 \approx 4.74$, while $|A_6| \approx 1.85$ and $|A_8| = 0$. Distances decide; compute the formula only to break ties.

Common trap

NumPy's `fft` uses $e^{-2\pi ixy/N}$ without normalisation, `ifft` uses $\frac{1}{N} e^{+2\pi ixy/N}$. With our sign, $\beta = \sqrt{N}\,$`ifft`(α) (equivalently `ifft(alpha, norm="ortho")`). `fft`$(\alpha)/\sqrt{N}$ is the *inverse* QFT.

4.6 Cost: FFT versus QFT

The classical FFT needs $\approx N \log N = n2^n$ operations; the QFT circuit needs $\frac{n(n+1)}{2} + \lfloor n/2 \rfloor$ gates. For $n = 10$: $10 \cdot 1024 = 10\,240$ versus $55 + 5 = 60$, a ratio ≈ 170. But the comparison is misleading: the QFT acts on amplitudes that cannot be read out (Section 10).

5 Qubits, states and measurement

5.1 States and the Born rule

A qubit state is $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. n qubits need 2^n amplitudes: $|\psi\rangle = \sum_x c_x\,|x\rangle$. **Born rule:** measuring all qubits gives x with probability $|c_x|^2$.

Method: Partial measurement

To measure only qubit j: (1) group the basis states by the value of bit j; (2) the probability of value b is the sum of $|c_x|^2$ over the group; (3) the post-measurement state is that group alone, *re-normalised*.

Worked example 5.11: Normalise, then measure one qubit

$|\psi\rangle = c(2\,|00\rangle - i\,|01\rangle + |10\rangle + \sqrt{3}\,|11\rangle)$, $c > 0$. Find $P(\text{qubit } 2 = 1)$ and the state afterwards.

Step 1. $c^2(4 + 1 + 1 + 3) = 9c^2 = 1 \Rightarrow c = \frac{1}{3}$.

Step 2. Qubit 2 is the *second* bit: group $\{|01\rangle, |11\rangle\}$. $P = \frac{1+3}{9} = \frac{4}{9}$.

Step 3. Post-state: $\frac{-i|01\rangle + \sqrt{3}|11\rangle}{2} = \frac{-i|0\rangle + \sqrt{3}|1\rangle}{2} \otimes |1\rangle$.

5.2 Global versus relative phase

Key result: Phases

$e^{i\gamma}|\psi\rangle$ and $|\psi\rangle$ are **the same physical state** (global phase): all probabilities in every basis agree. A **relative phase**, as in $|0\rangle + e^{i\phi}|1\rangle$, is physical: it changes results after an interfering gate such as H.

To test whether two vectors are the same state, divide one by the other component-wise: the ratio must be the *same* complex number for every component. Example: $(1, i)$ and $(-i, 1)$: ratios $-i/1 = -i$ and $1/i = -i$ — same state. $(1, i)$ and $(1, -i)$: ratios 1 and -1 — different states (they are orthogonal!).

5.3 The Bloch sphere

Every qubit state is (up to global phase) $\cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$, the point $(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. Then $\langle X\rangle = \sin\theta\cos\phi$, $\langle Y\rangle = \sin\theta\sin\phi$, $\langle Z\rangle = \cos\theta$. *Derivation:* with $a = \cos\frac{\theta}{2}$, $b = e^{i\phi}\sin\frac{\theta}{2}$: $\langle X\rangle = 2\mathrm{Re}(\bar{a}b)$, $\langle Y\rangle = 2\mathrm{Im}(\bar{a}b)$, $\langle Z\rangle = |a|^2 - |b|^2$, and $2\cos\frac{\theta}{2}\sin\frac{\theta}{2} = \sin\theta$, $\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = \cos\theta$. The QFT rotations R_k change only ϕ (rotation about z).

5.4 Superposition versus mixture

A coin that prepares $|0\rangle$ or $|1\rangle$ with probability $\frac{1}{2}$ each is a *mixture*: average the probabilities of the two cases, never add amplitudes. After H: each case gives $P(0) = \frac{1}{2}$, so $P(0) = \frac{1}{2}$. The superposition $|+\rangle$ gives $H|+\rangle = |0\rangle$, $P(0) = 1$. Interference needs coherence.

5.5 Repetition and confidence

If one run succeeds with probability p, then k independent runs succeed at least once with probability $1 - (1-p)^k$. To exceed $1 - \delta$ choose the smallest k with $(1-p)^k < \delta$. *Example:* $p = \frac{1}{2}$, $\delta = 0.01$: $2^{-k} < 0.01 \Rightarrow k = 7$ ($2^7 = 128$).

5.6 Memory of a state-vector simulation

2^n amplitudes $\times$ bytes per amplitude: complex single precision = 8 B, double = 16 B. $n = 30$ in single precision: $2^{30} \cdot 8$ B = 8 GiB; each extra qubit doubles it.

6 Quantum gates

6.1 The single-qubit gates of the QFT

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix}.$$

$H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$; compactly $H|x\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.x}|1\rangle)$. That last form is the seed of the whole QFT. Also $R_k^2 = R_{k-1}$, $R_k^\dagger = \mathrm{diag}(1, e^{-2\pi i/2^k})$.

Key result: H turns relative phase into probability

$H\frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}} = \frac{1+e^{i\phi}}{2}|0\rangle + \frac{1-e^{i\phi}}{2}|1\rangle$, so $P(0) = \cos^2\frac{\phi}{2}$, $P(1) = \sin^2\frac{\phi}{2}$.

(Use $|1 \pm e^{i\phi}|^2 = 2 \pm 2\cos\phi$.)

On n qubits: $H^{\otimes n}|x\rangle = \frac{1}{\sqrt{N}}\sum_y (-1)^{x\cdot y}|y\rangle$ with $x \cdot y = \sum_j x_j y_j$. For any $x \neq 0$ exactly half the signs are negative.

6.2 Controlled gates

Controlled-U applies U to the target only in the branch where the control is $|1\rangle$: $CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$. For a phase gate, only $|11\rangle$ is changed: $CR_k = \mathrm{diag}(1, 1, 1, e^{2\pi i/2^k})$ — **symmetric**: control and target can be swapped.

Worked example 6.12: Which gates commute with $CR_2 = \mathrm{diag}(1, 1, 1, i)$?

- $T \otimes T$: diagonal $\Rightarrow$ commutes.
- SWAP: exchanges $|01\rangle \leftrightarrow |10\rangle$, both have entry 1 $\Rightarrow$ commutes.
- CNOT (control 1): exchanges $|10\rangle \leftrightarrow |11\rangle$, entries 1 and i differ $\Rightarrow$ does *not* commute.
- $I \otimes H$: mixes $|10\rangle$ and $|11\rangle$ (entries 1, i) $\Rightarrow$ does not commute.

6.3 Phase kickback

Key result: Kickback

If $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, then $CU^m(|+\rangle|u\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, m\varphi}|1\rangle)|u\rangle$. The target is unchanged; the eigenvalue becomes a *relative* phase on the control.

Example: $X|-\rangle = -|-\rangle$, so CNOT$|+\rangle|-\rangle = |-\rangle|-\rangle$. *Example:* $U = T$, $|u\rangle = |1\rangle$, $\varphi = \frac{1}{8}$, $m = 2$: control becomes $\frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$.

6.4 Entanglement

Key result: Product test for two qubits

$\sum c_{jk}|jk\rangle$ is a product state $\iff$ $c_{00}c_{11} = c_{01}c_{10}$ (the 2×2 coefficient matrix has determinant 0).

Examples: $\frac{1}{2}(1, 1, 1, -1)$: $-1 \neq 1 \Rightarrow$ entangled (it is $CZ|+\rangle|+\rangle$). $\frac{1}{2}(1, -1, i, -i)$: $-i = -i \Rightarrow$ product $\frac{1}{2}(|0\rangle + i|1\rangle)(|0\rangle - |1\rangle)$. To factor, read the first row (c_{00}, c_{01}) as the second qubit's state and the ratio of rows as the first qubit's.

6.5 Bell states in another basis

Apply $H \otimes H$ term by term using $H^{\otimes 2}|x\rangle = \frac{1}{2}\sum_y (-1)^{x\cdot y}|y\rangle$. *Example:* $H^{\otimes 2}\frac{|01\rangle+|10\rangle}{\sqrt{2}}$: for $|01\rangle$ signs $(+, -, +, -)$, for $|10\rangle$ signs $(+, +, -, -)$; adding gives $\frac{1}{2\sqrt{2}}(2|00\rangle - 2|11\rangle) = \frac{|00\rangle-|11\rangle}{\sqrt{2}}$.

7 The QFT: definition and first calculations

Definition: Quantum Fourier transform

$\mathrm{QFT}\,|x\rangle = \frac{1}{\sqrt{N}}\sum_{y=0}^{N-1}\omega^{xy}\,|y\rangle$, extended linearly: $\mathrm{QFT}\sum_x \alpha_x\,|x\rangle = \sum_y \beta_y\,|y\rangle$ with $\beta = F\alpha$. It is the DFT of Section 4 applied to the amplitude vector.

Method: Amplitude of $|y\rangle$ in $\mathrm{QFT}\,|x\rangle$

(1) Convert both bit strings to integers. (2) Compute $xy \bmod N$. (3) Amplitude $= \omega^{xy}/\sqrt{N}$; phase in degrees $= 360^\circ \cdot \frac{xy \bmod N}{N}$.

Worked example 7.13: Three quick amplitudes

1. $n = 2$, $x = |10\rangle = 2$, $y = |11\rangle = 3$: $\omega = i$, $i^6 = -1$, amplitude $-\frac{1}{2}$.
2. $n = 3$, $x = 5$, $y = 6$: $30 \bmod 8 = 6$, $\omega^6 = e^{2\pi i \cdot 6/8} = -i$, amplitude $-\frac{i}{\sqrt{8}}$; phase 270°.
3. $n = 4$, $x = 7$, $y = 3$: $21 \bmod 16 = 5$, phase $360^\circ \cdot \frac{5}{16} = 112.5^\circ$.

Worked example 7.14: Full output for $n = 2$

$\mathrm{QFT}\,|01\rangle = \frac{1}{2}(|0\rangle + i\,|1\rangle + i^2\,|2\rangle + i^3\,|3\rangle) = \frac{1}{2}(|00\rangle + i\,|01\rangle - |10\rangle - i\,|11\rangle)$. Probability of each outcome $\frac{1}{4}$: the information about x is entirely in the phases.

Key result: Special inputs

- $\mathrm{QFT}\,|0\rangle = \frac{1}{\sqrt{N}}\sum_y |y\rangle = |+\rangle^{\otimes n}$ (uniform).
- $\mathrm{QFT}\big(\frac{1}{\sqrt{N}}\sum_x |x\rangle\big) = |0\rangle$ (orthogonality).
- $\mathrm{QFT}\,|2^{n-1}\rangle = \mathrm{QFT}\,|10\ldots0\rangle$: amplitudes $\frac{(-1)^y}{\sqrt{N}}$ depend only on the last bit y_n: $|+\rangle^{\otimes(n-1)}\,|-\rangle$.
- $\mathrm{QFT}\,|1\rangle = \mathrm{QFT}\,|0\ldots01\rangle$: amplitudes ω^y: the slowest "rotating" signal.

8 The product formula

This is the single most useful result about the QFT.

Key result: Product formula

$$\mathrm{QFT}\,|x_1\ldots x_n\rangle = \frac{1}{\sqrt{N}}\big(\,|0\rangle + e^{2\pi i\,0.x_n}\,|1\rangle\,\big)\big(\,|0\rangle + e^{2\pi i\,0.x_{n-1}x_n}\,|1\rangle\,\big)\cdots\big(\,|0\rangle + e^{2\pi i\,0.x_1x_2\ldots x_n}\,|1\rangle\,\big).$$

Output qubit l (counting from the left) carries the phase $0.x_{n-l+1}\ldots x_n$: the last l bits of x.

Derivation.

Step 1. Write $y = \sum_{l=1}^{n} y_l 2^{n-l}$. Then $\frac{xy}{2^n} = \sum_l \frac{xy_l}{2^l}$, so $\omega^{xy} = \prod_{l=1}^{n} e^{2\pi i\, xy_l/2^l}$.

Step 2. The sum over y is a sum over independent bits, so it factorises:

$$\sum_{y_1,\dots,y_n} \prod_l e^{2\pi i\, x y_l/2^l} |y_l\rangle = \prod_l \sum_{y_l\in\{0,1\}} e^{2\pi i\, x y_l/2^l} |y_l\rangle = \prod_l (\,|0\rangle + e^{2\pi i\, x/2^l} |1\rangle\,).$$

Step 3. By Section 2, $e^{2\pi i\, x/2^l} = e^{2\pi i\, 0.x_{n-l+1}\dots x_n}$.

Worked example 8.15: QFT $|101\rangle$ with the product formula

$x_1x_2x_3 = 101$.

Step 1. Qubit 1: $0.x_3 = 0.1_2 = \frac{1}{2}$: $e^{i\pi} = -1$: $(|0\rangle - |1\rangle)$.

Step 2. Qubit 2: $0.x_2x_3 = 0.01_2 = \frac{1}{4}$: $e^{i\pi/2} = i$: $(|0\rangle + i\,|1\rangle)$.

Step 3. Qubit 3: $0.x_1x_2x_3 = 0.101_2 = \frac{5}{8}$: $e^{5i\pi/4}$: $(|0\rangle + e^{5i\pi/4}\,|1\rangle)$.

QFT $|101\rangle = \frac{1}{\sqrt{8}}(|0\rangle - |1\rangle)(|0\rangle + i\,|1\rangle)(|0\rangle + e^{5i\pi/4}\,|1\rangle)$. *Check one amplitude:* $y = |110\rangle = 6$: product formula gives $(-1)(i)(1) = -i$; definition gives $\omega^{30} = \omega^6 = -i$. ✓

Key result: Consequence

QFT $|x\rangle$ is **always a product state** for a basis input x — never entangled. (Superposition inputs can give entangled outputs.)

Common trap

Writing the factors in reverse order (qubit 1 with phase $0.x_1\dots x_n$) is exactly the state *before* the final SWAPs of the circuit. Always check one amplitude with the definition.

9 The QFT circuit

9.1 From the product formula to gates

Look at the last factor $|0\rangle + e^{2\pi i\, 0.x_1x_2\dots x_n}\,|1\rangle$. Build it on qubit 1:

Step 1. H on qubit 1: $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.x_1}\,|1\rangle)$, because $(-1)^{x_1} = e^{2\pi i\, x_1/2}$.

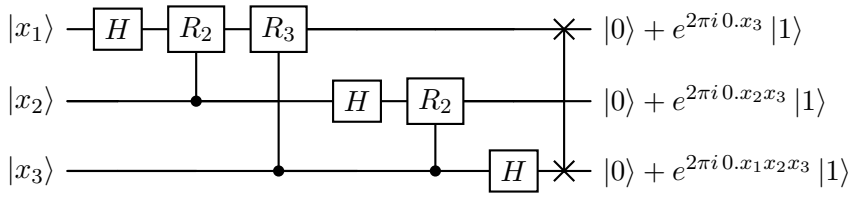
Step 2. Controlled-R_2 from qubit 2: if $x_2 = 1$ multiply the $|1\rangle$ part by $e^{2\pi i/4}$: phase becomes $0.x_1x_2$.

Step 3. Controlled-R_k from qubit k appends bit x_k at position k; after all of them: $0.x_1x_2\dots x_n$.

Step 4. Repeat for qubit 2 (using qubits $3,\dots,n$), and so on. Qubit j ends with $0.x_j\dots x_n$.

Step 5. The order is reversed with respect to the product formula, so finish with SWAPs of qubits $j \leftrightarrow n+1-j$.

Why must qubit 1 be processed first? Its rotations are controlled by qubits $2,\dots,n$, which must still hold the original bits $x_2,\dots,x_n$; qubit 2 is only changed afterwards.



The 3-qubit textbook QFT (normalisation $\frac{1}{\sqrt{8}}$ omitted on the right).

9.2 Gate counts

Key result: Counting

Hadamards: n. Controlled rotations: $(n-1)+(n-2)+\cdots+0=\frac{n(n-1)}{2}$. SWAPs: $\lfloor n/2 \rfloor$. Total gates $\frac{n(n+1)}{2}+\lfloor n/2 \rfloor$. Gate R_k appears $n-k+1$ times ($k=2,\ldots,n$); the smallest angle is $\frac{360^\circ}{2^n}$ (gate R_n).

Worked example 9.16: Counting for $n=9$

CR: $\frac{9\cdot 8}{2}=36$. SWAPs: 4. Two-qubit gates if a SWAP costs 3 CNOTs: $36+12=48$. Smallest angle $360^\circ/512\approx 0.703^\circ$. Number of R_3 gates: $9-3+1=7$.

9.3 The inverse QFT

Method: Inverting any circuit

$(G_m\cdots G_2G_1)^{-1}=G_1^{-1}G_2^{-1}\cdots G_m^{-1}$: **reverse the order** and **invert each gate**. $H^{-1}=H$, $\text{SWAP}^{-1}=\text{SWAP}$, $R_k^{-1}=R_k^\dagger$ (negative angle).

For 3 qubits the QFT applies: $H_1, CR_2^{(2\to1)}, CR_3^{(3\to1)}, H_2, CR_2^{(3\to2)}, H_3, \text{SWAP}_{13}$. So QFT^{-1} applies: SWAP_{13}, H_3, $CR_2^{\dagger(3\to2)}$, H_2, $CR_3^{\dagger(3\to1)}$, $CR_2^{\dagger(2\to1)}$, H_1. The first non-SWAP gate is H on the *last* qubit.

9.4 Tracing the circuit gate by gate

Worked example 9.17: Input $|110\rangle$, phases in degrees

$x_1=1$, $x_2=1$, $x_3=0$.

Step 1. H_1: phase $180^\circ\cdot x_1=180^\circ$. CR_2 (control $x_2=1$): $+90^\circ\to 270^\circ$. CR_3 (control $x_3=0$): unchanged, 270°. Check: $0.110_2=\frac{3}{4}\to 270^\circ$. ✓

Step 2. H_2: $180^\circ\cdot x_2=180^\circ$. CR_2 from $x_3=0$: unchanged. Qubit 2: 180° ($0.10_2=\frac{1}{2}$).

Step 3. H_3: $x_3=0\to 0^\circ$.

Step 4. SWAP_{13}: qubit 1 now has 0°, qubit 2 180°, qubit 3 270°.

Step 5. Amplitude of, say, $|011\rangle$: sum of phases of the qubits that are 1: $180^\circ+270^\circ=450^\circ\equiv 90^\circ$, so $\frac{i}{\sqrt{8}}$. Definition: $x=6$, $y=3$, $18 \bmod 8=2$, $\omega^2=i$. ✓

Without the SWAP the same register $|011\rangle$ would collect $180^\circ+0^\circ=180^\circ$ — a different (wrong) answer.

10 Algebraic properties of the QFT

Key result: QFT^2 and QFT^4

$\mathrm{QFT}^2 |x\rangle = |-x \bmod N\rangle$ and therefore $\mathrm{QFT}^4 = I$.

Derivation: $\mathrm{QFT}^2 |x\rangle = \frac{1}{N} \sum_z \left(\sum_y \omega^{y(x+z)} \right) |z\rangle$; the inner sum is $N\delta_{z,-x}$. Not to be confused with the bit complement $|N-1-x\rangle$: for $x=0$, $\mathrm{QFT}^2 |0\rangle = |0\rangle$ but the complement is $|N-1\rangle$.

Eigenvalues. From $\mathrm{QFT}^4 = I$ every eigenvalue satisfies $\lambda^4 = 1$: $\lambda \in \{1, -1, i, -i\}$. Use trace and determinant to see which occur. For $N=4$: $\operatorname{tr} F_4 = \frac{1}{2} \sum_x i^{x^2} = \frac{1}{2}(1+i+1+i) = 1+i$ and $\det F_4 = -i$; eigenvalues $1, 1, -1, i$ (sum $1+i$ ✓, product $-i$ ✓). For $N=2$: H has ± 1. For $N \geq 8$ all four values occur. So $\det F_N$ need not be ± 1 although $|\det F_N| = 1$.

Other matrices: F_4^2 is the permutation $0 \to 0$, $1 \leftrightarrow 3$, $2 \to 2$; $H \otimes H$ is real, symmetric and self-inverse but not a permutation; CR_k is diagonal with $\det = e^{2\pi i/2^k}$.

11 What the QFT can and cannot do

Key result: Measuring right after a QFT on a basis state

Every outcome has probability $\frac{1}{N}$, independent of x: **no information**. The reason is that all amplitudes have the same magnitude $\frac{1}{\sqrt{N}}$.

- **Fast:** $O(n^2)$ gates transform 2^n amplitudes.
- **Not a fast classical FFT:** loading an arbitrary vector costs up to $\Omega(2^n)$ gates, and a measurement returns a single y with probability $|\beta_y|^2$; estimating all coefficients needs exponentially many runs.
- **Useful as a subroutine** when only a few large peaks must be found: phase estimation, period finding, Shor.

Assertion–Reason tip: "all magnitudes equal" *explains* "no information"; "$O(n^2)$ gates" does *not* imply "all coefficients quickly".

12 Quantum phase estimation

12.1 One counting qubit: the Hadamard test

Circuit: control in $|0\rangle$, H, controlled-U on eigenstate $|u\rangle$ ($U|u\rangle = e^{2\pi i\varphi}|u\rangle$), H, measure.

Step 1. After H and kickback: $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\varphi}|1\rangle)|u\rangle$.

Step 2. After the second H (Section 6, $\phi = 2\pi\varphi$): $P(0) = \cos^2(\pi\varphi)$, $P(1) = \sin^2(\pi\varphi)$.

Example: $\varphi = \frac{1}{4}$: $P(1) = \sin^2 \frac{\pi}{4} = \frac{1}{2}$; $\varphi = \frac{1}{6}$: $P(1) = \frac{1}{4}$.

12.2 t counting qubits

Counting qubit j controls $U^{2^{t-j}}$. By kickback the register becomes

$$\frac{1}{\sqrt{2^t}} \sum_{k=0}^{2^t-1} e^{2\pi i \varphi k} |k\rangle .$$

If $\varphi = \frac{b}{2^t}$ exactly, this is QFT $|b\rangle$, so the inverse QFT returns $|b\rangle$ **with certainty**.

Key result: Outcome distribution in general

With $\delta = \varphi - \frac{y}{2^t}$:

$$P(y) = \frac{1}{2^{2t}} \left| \sum_{k=0}^{2^t-1} e^{2\pi i k \delta} \right|^2 = \frac{\sin^2(\pi 2^t \delta)}{2^{2t} \sin^2(\pi \delta)}.$$

Worked example 12.18: $t = 2$, $\varphi = \frac{1}{5}$

Step 1. Grid points $y/4$: $0, 0.25, 0.5, 0.75$. Closest to 0.2 is $y = 1$, $\delta = 0.2 - 0.25 = -0.05$.

Step 2. $P(1) = \frac{\sin^2(4\pi \cdot 0.05)}{16 \sin^2(\pi \cdot 0.05)} = \frac{\sin^2(0.2\pi)}{16 \sin^2(0.05\pi)} = \frac{0.3455}{16 \cdot 0.02447} \approx 0.883$.

Step 3. $y = 0$: $\delta = 0.2$: $\frac{\sin^2(0.8\pi)}{16 \sin^2(0.2\pi)} = \frac{0.3455}{16 \cdot 0.3455} = \frac{1}{16}$.

Exact surds: write $\sin^2 a = \frac{1 - \cos 2a}{2}$ to simplify, e.g. $\sin^2 \frac{\pi}{12} = \frac{2 - \sqrt{3}}{4}$, then rationalise.

12.3 The $4/\pi^2$ guarantee

For the closest grid point $|\delta| \le 2^{-(t+1)}$, so $x = \pi 2^t |\delta| \le \frac{\pi}{2}$. Use $\sin x \ge \frac{2x}{\pi}$ on $[0, \frac{\pi}{2}]$ and $\sin \pi|\delta| \le \pi|\delta|$:

$$\sqrt{P} \ge \frac{2 \cdot 2^t |\delta|}{2^t \pi |\delta|} = \frac{2}{\pi} \quad \Rightarrow \quad P(\text{closest}) \ge \frac{4}{\pi^2} \approx 0.405.$$

The probability of getting one of the *two* closest grid points is at least $\frac{8}{\pi^2} \approx 0.81$.

12.4 How many counting qubits?

To get n correct bits with probability $\ge 1 - \varepsilon$: $t = n + \lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$. *Example:* $n = 4$, $\varepsilon = 0.05$: $2 + 10 = 12$, $\lceil \log_2 12 \rceil = 4$, $t = 8$.

Common trap

If φ has no terminating binary expansion (e.g. $\frac{1}{3}$), no finite t gives it exactly; the outcome is always an integer y and $y/2^t \ne \varphi$. The distribution is only *peaked* near $2^t \varphi$.

13 Period finding and Shor's algorithm

13.1 From factoring to order finding

For N odd composite and a coprime to N, the **order** r is the least $r > 0$ with $a^r \equiv 1 \pmod N$. If r is even and $a^{r/2} \not\equiv -1$, then $\gcd(a^{r/2} \pm 1, N)$ are non-trivial factors.

Worked example 13.19: Classical part for $N = 33$, $a = 5$

Powers of 5 mod 33: $5, 25, 26, 31, 23, 16, 14, 4, 20, 1$, so $r = 10$. $5^5 \equiv 23 \not\equiv 32$. $\gcd(22, 33) = 11$, $\gcd(24, 33) = 3$. $33 = 3 \times 11$. ✓ Tip: r divides $\varphi(N)$ ($\varphi(33) = 20$), which limits the candidates.

13.2 The quantum part

Registers: t counting qubits (usually $t = 2n$ with $n = \lceil \log_2 N \rceil$) and n work qubits: total $3n$ qubits.

Step 1. $H^{\otimes t}$, then $|x\rangle\,|1\rangle \mapsto |x\rangle\,|a^x \bmod N\rangle$: $\frac{1}{\sqrt{M}} \sum_x |x\rangle\,|a^x\rangle$, $M = 2^t$.

Step 2. Group by work value: $x = x_0 + rj$. Different work values are orthogonal, so each branch x_0 is a periodic state in the counting register (Section 4) and they do not interfere.

Step 3. Inverse QFT: peaks near $y \approx kM/r$, $k = 0, \ldots, r-1$. If $r \mid M$ the peaks are exact with probability $\frac{1}{r}$ each.

Step 4. Continued fractions turn y/M into k/r.

13.3 Continued fractions

Method: Expanding y/M

Repeat: write the integer part, invert the remainder. Convergents h_j/k_j from $h_j = a_j h_{j-1} + h_{j-2}$, $k_j = a_j k_{j-1} + k_{j-2}$ ($h_{-1} = 1, h_{-2} = 0, k_{-1} = 0, k_{-2} = 1$). Keep the last convergent with denominator $< N$.

Worked example 13.20: Continued fractions for $N = 33$, $a = 5$ ($r = 10$), $M = 2048$

The peaks are near $y \approx k \cdot \frac{2048}{10} = 204.8k$.

Step 1. $k = 3$: measured $y = 614$. $\frac{614}{2048}$: $2048 = 3 \cdot 614 + 206$; $614 = 2 \cdot 206 + 202$; $206 = 1 \cdot 202 + 4$; $202 = 50 \cdot 4 + 2$; $4 = 2 \cdot 2$. So $[0; 3, 2, 1, 50, 2]$ with convergents $\frac{1}{3}, \frac{2}{7}, \frac{3}{10}, \frac{152}{507}, \ldots$ The last with denominator < 33 is $\frac{3}{10}$: $r = 10$, and $5^{10} \equiv 1 \pmod{33}$ ✓.

Step 2. $k = 2$: measured $y = 410$. $\frac{410}{2048} = [0; 4, 1, 204]$, convergents $\frac{1}{4}, \frac{1}{5}, \frac{205}{1024}$: candidate $r' = 5$. But $5^5 \equiv 23 \not\equiv 1$: rejected. Because $\gcd(2, 10) = 2$ the fraction $\frac{2}{10}$ reduced to $\frac{1}{5}$.

Step 3. $k = 6$: $y = 1229$: $[0; 1, 1, 1, 1, 409]$, convergents $1, \frac{1}{2}, \frac{2}{3}, \frac{3}{5}$: again $r' = 5$ (divisor). $\mathrm{lcm}(5, \ldots)$ with a later candidate 2 would give 10.

Two real phenomena: (1) outcomes far from a peak give garbage, rejected by checking $a^{r'} \equiv 1$; (2) when $\gcd(k, r) > 1$ (above: $k = 2, 6$ with $r = 10$) the fraction reduces and only a **divisor** $r/\gcd(k, r)$ appears. Fix: repeat and take the lcm of the candidates, or test small multiples. The fraction of good k is $\varphi(r)/r$.

Worked example 13.21: An exact case: $a = 2$, $N = 15$, $t = 8$

Order: $2, 4, 8, 1 \Rightarrow r = 4$, and $4 \mid 256$. Outcomes $\{0, 64, 128, 192\}$, each $\frac{1}{4}$. Fractions: $0 \to$ no information; $64/256 = \frac{1}{4} \to 4$ ✓; $128/256 = \frac{1}{2} \to 2$ (divisor, $2^2 = 4 \not\equiv 1$, rejected); $192/256 = \frac{3}{4} \to 4$ ✓.

13.4 Peak probability when $r \nmid M$

Count the terms in each branch: $M = qr + s$ $(0 \le s < r)$: s branches have $q+1$ terms, $r-s$ have q. At an exact grid point y with ry/M an integer every phase is 1 and $P(y) = \frac{\sum_{x_0} K_{x_0}^2}{M^2}$. *Example:* $r = 6$, $M = 64$: $64 = 10 \cdot 6 + 4$: four branches with 11, two with 10. $y = 32$ $(6 \cdot 32/64 = 3)$: $P = \frac{4 \cdot 121 + 2 \cdot 100}{4096} = \frac{684}{4096} \approx 0.167 \approx \frac{1}{6}$.

14 Approximate QFT, semiclassical QFT and applications

14.1 Approximate QFT

Drop every CR_k with $k > m$. Kept gates: $\sum_{k=2}^{m}(n-k+1)$; dropped: $\frac{n(n-1)}{2} -$ kept. *Example:* $n = 12$, $m = 4$: kept $11 + 10 + 9 = 30$, total 66, dropped 36. **Error bound:** $\|CR_k - I\| = |e^{2\pi i/2^k} - 1| \le \frac{2\pi}{2^k}$; errors of a product of unitaries add, so the total error is at most $\sum_{k>m} n\frac{2\pi}{2^k} < \frac{2\pi n}{2^m}$. With $m = \log_2 n + c$ the error is $\le 2\pi/2^c$, independent of n.

14.2 Semiclassical QFT (measure as you go)

When the inverse QFT is immediately followed by measurement, measure each qubit as soon as its last gate is done. Because controlled-phase gates are symmetric, the already-measured qubit can be the control; a control that is a classical bit means "apply $R_k^\dagger$ if the bit is 1". **No two-qubit gates remain**; only measurements and classically controlled single-qubit rotations.

14.3 Where the QFT is used

Algorithm	Role of the QFT
Phase estimation	inverse QFT converts phases $e^{2\pi i 2^j \varphi}$ into the binary digits of φ
Shor / order finding	phase estimation of $\lvert x\rangle \mapsto \lvert ax \bmod N\rangle$; continued fractions give r
Draper adder	in the Fourier basis, adding a is the phase ramp ω^{ay} (shift theorem): controlled rotations, then QFT^{-1}
HHL linear solver	phase estimation writes eigenvalues λ; an ancilla rotation $\propto 1/\lambda$ performs A^{-1}
Hidden subgroup problems	QFT over the group reveals the subgroup's "frequencies"

15 Problem-solving strategies

15.1 Strategy by question format

- **Single-correct numerical:** compute fully, then sanity-check (probabilities in $[0,1]$, sum to 1; phases mod 360°; Parseval). Many distractors are "forgot normalisation", "amplitude instead of probability", "wrong sign of exponent", "bit order reversed".
- **Matching lists:** solve the easiest row first; often one row fixes the option. Eliminate options as you go.
- **Assertion–Reason:** decide A and R *independently* first. Only if both are true ask: does R *cause* A? Watch for over-general reasons ("always distinct", "global phase").
- **Statement I/II:** judge each statement alone; one counterexample kills a "for every" statement.

- **Paragraph:** the first question usually sets up a quantity reused in the second; write it down clearly.

15.2 Fast checks that catch most mistakes

1. Is the state normalised? Is the matrix unitary?
2. For a QFT amplitude: $|\text{amplitude}| = 1/\sqrt{N}$ for basis inputs.
3. Compute one amplitude both from the definition and from the product formula / circuit.
4. Is the period dividing N? If yes, exact peaks $\frac{1}{r}$; if no, leakage.
5. Does $\gcd(k, r) > 1$? Then continued fractions give a divisor.

15.3 Map: assessment questions → sections of this chapter

Questions	Where to learn it
Q.1–Q.6, Q.12, Q.15, Q.17	Section 1 (roots of unity, geometric series, products, peak shape, Gauss sums)
Q.3, Q.14, Q.78	Section 2 (binary fractions)
Q.4, Q.5, Q.16, Q.18, Q.30, Q.31, Q.73	Sections 3 and 10 (unitary matrices, determinants, eigenvalues)
Q.11, Q.13, Q.19, Q.20, Q.53, Q.63, Q.72	Section 4 (DFT, Parseval, shift theorem, leakage, NumPy)
Q.7, Q.8, Q.68–Q.70, Q.79, Q.80	Section 13 (orders, continued fractions, Shor)
Q.9, Q.21–Q.24, Q.29, Q.32, Q.34, Q.37, Q.40	Section 5 (states, measurement, phases, mixtures, memory)
Q.10, Q.25–Q.28, Q.33, Q.35, Q.36, Q.38	Section 6 (gates, kickback, entanglement)
Q.39	Section 12 (Hadamard test)
Q.41–Q.43, Q.49, Q.50, Q.52, Q.54, Q.56	Sections 7–8 (definition, product formula)
Q.44–Q.47, Q.51, Q.55, Q.59, Q.60	Section 9 (circuit, counts, inverse, gate-by-gate tracing)
Q.48, Q.57, Q.58	Sections 10–11 (algebra, limits of the QFT)
Q.61, Q.62, Q.76	Section 4 (periodic states)
Q.64–Q.67	Section 12 (phase estimation)
Q.71, Q.74, Q.75, Q.77	Section 14 (approximate / semiclassical QFT, applications)

(Section numbers: 1 roots of unity, 2 binary, 3 linear algebra, 4 DFT, 5 states, 6 gates, 7 QFT definition, 8 product formula, 9 circuit, 10 algebraic properties, 11 limits, 12 phase estimation, 13 Shor, 14 approximate QFT.)

16 Practice problems

Try each before looking at the answers on the next page. They use the same ideas as the assessment but different numbers.

P1. Compute $\sum_{k=0}^{5} e^{2\pi i\, 3k/6}$ and $\sum_{k=0}^{5} e^{2\pi i\, 6k/6}$.

P2. Find $e^{2\pi i\, x/2^3}$ for $x = 53 = 110101_2$.

P3. Is $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & i \\ 1 & -i \end{smallmatrix}\right)$ unitary?

P4. Give $|\beta_0|^2$ for the DFT ($N = 4$) of $\alpha = \frac{1}{\sqrt{6}}(1, 1, 2, 0)$.

P5. $|\psi\rangle = \frac{1}{\sqrt{6}}(|00\rangle + 2\,|01\rangle + i\,|11\rangle)$. Find $P(\text{qubit } 1 = 0)$ and the state of qubit 2 afterwards.

P6. Is $\frac{1}{2}(|00\rangle + i\,|01\rangle + i\,|10\rangle - |11\rangle)$ entangled? If not, factor it.

P7. For $n = 3$ write $\mathrm{QFT}\,|011\rangle$ with the product formula and give the amplitude of $|100\rangle$.

P8. Number of controlled rotations, SWAPs and R_4 gates in the QFT on $n = 7$ qubits.

P9. Phase (degrees) of the amplitude of $|y = 9\rangle$ in $\mathrm{QFT}\,|x = 3\rangle$ for $n = 4$.

P10. $\mathrm{QFT}^2\,|5\rangle$ for $n = 3$.

P11. The state $\frac{1}{2}(|2\rangle + |6\rangle + |10\rangle + |14\rangle)$ ($n = 4$) is Fourier transformed. Outcomes and probabilities?

P12. QPE with $t = 3$ and $\varphi = \frac{3}{8}$: which outcome, with which probability?

P13. QPE with $t = 2$, $\varphi = \frac{1}{6}$: $P(y = 1)$ exactly.

P14. Order of 4 modulo 21, and what does Shor's gcd step give?

P15. $N = 15$, $a = 4$, $t = 8$: possible outcomes y.

P16. Approximate QFT, $n = 16$, keep $k \le 5$: number of dropped controlled rotations.

Answers to the practice problems

P1. $6 \nmid 3$: sum 0 (it is $1-1+1-1+1-1$). $6 \mid 6$: sum 6.

P2. $53/8 = 110.101_2$, fractional part $0.101_2 = \frac{5}{8}$: $e^{5\pi i/4}$.

P3. Columns $\frac{1}{\sqrt{2}}(1,1)$ and $\frac{1}{\sqrt{2}}(i,-i)$: inner product $\frac{1}{2}(i-i)=0$, norms 1. Yes.

P4. $\beta_0 = \frac{1}{2}\cdot\frac{4}{\sqrt{6}}$, $|\beta_0|^2 = \frac{4}{6} = \frac{2}{3}$.

P5. Group $\{|00\rangle, |01\rangle\}$: $P = \frac{1+4}{6} = \frac{5}{6}$; qubit 2 becomes $\frac{|0\rangle+2|1\rangle}{\sqrt{5}}$.

P6. $c_{00}c_{11} = -1$, $c_{01}c_{10} = i\cdot i = -1$: product. $\frac{1}{2}(|0\rangle + i|1\rangle)(|0\rangle + i|1\rangle)$.

P7. Phases $0.x_3 = 0.1 \to -1$; $0.x_2x_3 = 0.11 \to -i$; $0.011 = \frac{3}{8} \to e^{3\pi i/4}$: $\frac{1}{\sqrt{8}}(|0\rangle - |1\rangle)(|0\rangle - i|1\rangle)(|0\rangle + e^{3\pi i/4}|1\rangle)$. Amplitude of $|100\rangle$: $-\frac{1}{\sqrt{8}}$ (check: $\omega^{12} = \omega^4 = -1$).

P8. 21, 3, and $7-4+1=4$.

P9. $27 \bmod 16 = 11$: $360^\circ \cdot \frac{11}{16} = 247.5^\circ$.

P10. $|-5 \bmod 8\rangle = |3\rangle$.

P11. Period 4, $4 \mid 16$: $y \in \{0,4,8,12\}$, each $\frac{1}{4}$.

P12. $\frac{3}{8} = 0.011_2$ exactly: outcome 011 with probability 1.

P13. $\delta = \frac{1}{6} - \frac{1}{4} = -\frac{1}{12}$: $P = \frac{\sin^2(\pi/3)}{16\sin^2(\pi/12)} = \frac{3/4}{16(2-\sqrt{3})/4} = \frac{3(2+\sqrt{3})}{16} = \frac{6+3\sqrt{3}}{16} \approx 0.70$.

P14. $4, 16, 1$: $r = 3$, odd — the method fails for this a; choose another a.

P15. $4^2 = 16 \equiv 1$: $r = 2$; outcomes $\{0, 128\}$, each $\frac{1}{2}$.

P16. Total $\frac{16\cdot 15}{2} = 120$; kept $15+14+13+12 = 54$; dropped 66.

Summary of the chapter in ten lines

1. $\sum_j \omega^{jk} = N$ if $N \mid k$, else 0; whole turns vanish.
2. $e^{2\pi i x/2^l}$ keeps only the last l bits of x.
3. Unitary $\Leftrightarrow$ orthonormal columns; $|\det| = 1$; $|\lambda| = 1$.
4. Global phase is invisible; relative phase becomes probability after H: $P(0) = \cos^2(\phi/2)$.
5. Kickback: CU^m writes $e^{2\pi i m\varphi}$ on the control.
6. $\mathrm{QFT}|x\rangle = \frac{1}{\sqrt{N}}\sum_y \omega^{xy}|y\rangle$ = product of $(|0\rangle + e^{2\pi i\, 0.x_{n-l+1}\ldots x_n}|1\rangle)$.
7. Circuit: H + controlled R_k + SWAPs; $\frac{n(n-1)}{2}$ rotations; invert by reversing and conjugating.
8. $\mathrm{QFT}^2|x\rangle = |-x\rangle$, $\mathrm{QFT}^4 = I$, eigenvalues in $\{\pm 1, \pm i\}$.
9. Period $r \mid N$: r peaks spaced N/r, each $\frac{1}{r}$; otherwise leakage.
10. QPE: exact if $\varphi = b/2^t$, else $\geq 4/\pi^2$ for the closest estimate; Shor = QPE + continued fractions + gcd.